

# Why Machine Learning is Pervasive

## The Limits of Explicit Programming

- Machine Learning grew as a sub-field of Artificial Intelligence (AI) because many complex, real-world problems proved too difficult to solve by writing explicit, rule-based programs.
  - **Core Idea:** For many tasks, the only viable solution was to create systems that could **learn from data** on their own.
  - **Examples of such tasks include:**
    - Web search ranking
    - Human speech recognition
    - Diagnosing diseases from X-rays
    - Building self-driving cars
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## The Goal of This Course

- You will learn the most important **state-of-the-art algorithms**.
  - You will gain crucial **practical tips and tricks** for making these algorithms perform well.
  - You will get hands-on experience by **implementing the algorithms** yourself.
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## Future & Economic Impact 🚀

- Machine learning is expected to significantly impact nearly every industry, extending far beyond the software sector into areas like manufacturing, agriculture, and healthcare.
- **Artificial General Intelligence (AGI):** This is the long-term dream of building machines with human-level intelligence. While it is still a distant goal, learning algorithms are seen as the most promising path toward it.
- **Economic Value:** A McKinsey study estimates that AI and machine learning will create an additional **\$13 trillion** in value annually by the year 2030.
- **Career Opportunities:** Due to the massive number of untapped applications, there is a vast and unfulfilled demand for people with machine learning skills.